



Signifor® LAR

DESCRIPTION AND COMPOSITION

Signifor LAR is a long-acting depot injection form of pasireotide, powder for suspension for injection to be suspended in a solvent immediately prior to i.m. injection.

Pharmaceutical form(s)

Powder and solvent for suspension for injection

Powder: Slightly yellowish to yellowish powder in vial

Solvent for suspension for injection: clear, colorless to slightly yellow to slightly brown solution in pre-filled syringe

Active substance(s)

Each vial contains:

Signifor LAR 20 mg – 20 mg pasireotide (as pamotate)

Signifor LAR 40 mg - 40 mg pasireotide (as pamotate)

Signifor LAR 60 mg - 60 mg pasireotide (as pamotate)

CLINICAL PARTICULARS

Therapeutic indications

Signifor LAR is indicated for the treatment of adult patients with acromegaly for whom surgery is not an option or has not been curative and who are inadequately controlled on treatment with another somatostatin analogue.

Posology and method of administration

Posology

The recommended initial dose is 40 mg of pasireotide every 4 weeks.

The dose may be increased to a maximum of 60 mg for patients whose growth hormone (GH) and/or insulin like growth factor-1 (IGF-1) levels are not fully controlled after 3 months of treatment with Signifor LAR at 40 mg.

Management of suspected adverse reactions or over-response to treatment (IGF-1 < lower limit of normal) may require temporary dose reduction of Signifor LAR. The dose may be decreased either temporarily or permanently by 20 mg decrements.

If a dose of Signifor LAR is missed the missed injection should be administered as soon as possible. The next dose should then be planned for 4 weeks after the injection is administered in order to resume the normal schedule of one dose every 4 weeks.

Special populations

Elderly patients (≥65 years)

Data on the use of Signifor LAR in patients older than 65 years are limited, but there is no evidence to suggest that dose adjustment is required in these patients (see section Pharmacokinetic properties).

Renal impairment

No dose adjustment is required in patients with impaired renal function (see section Pharmacokinetic properties).

Hepatic impairment

Dose adjustment is not required in patients with mildly impaired hepatic function (Child Pugh A).

The recommended initial dose for patients with moderate hepatic impairment (Child Pugh B) is 20 mg every 4 weeks, and the maximum recommended dose for these patients is 40 mg every 4 weeks (see section Pharmacokinetic properties).

Signifor LAR should not be used in patients with severe hepatic impairment (Child Pugh C) (see sections Contraindications and Special warnings and precautions for use).

Paediatric population

The safety and efficacy of Signifor LAR in children and adolescents aged 0 to 18 years have not been established. No data are available.

Method of administration

Signifor LAR is to be administered by deep intramuscular injection by a trained healthcare professional.

Signifor LAR suspension must only be prepared immediately before administration.

The site of repeat intramuscular injections should be alternated between the left and right gluteal muscle.

For instructions on reconstitution of the medicinal product before administration (see section Instructions For Use).

Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section List of excipients.

Severe hepatic impairment (Child Pugh C).

Special warnings and precautions for use

Glucose metabolism

Alterations in blood glucose levels have been frequently reported in healthy volunteers and patients treated with pasireotide. Hyperglycaemia and, less frequently, hypoglycaemia, were observed in subjects participating in clinical studies with pasireotide (see section Undesirable effects).

In acromegaly patients who developed hyperglycaemia, the condition generally appeared to respond to antidiabetic therapy. Dose reductions or discontinuation of treatment with pasireotide due to hyperglycaemia were infrequent in clinical studies with pasireotide.

The development of hyperglycaemia appears to be related to decreases in secretion of insulin and of incretin hormones (i.e. glucagon-like peptide-1 [GLP-1] and glucose-dependent insulinotropic polypeptide [GIP]).

Glycaemic status (fasting plasma glucose/haemoglobin A_{1c} [FPG/HbA_{1c}]) should be assessed prior to starting treatment with pasireotide. FPG/HbA_{1c} monitoring during treatment should follow established guidelines. Self monitoring of blood glucose and/or FPG assessments should be done weekly for the first three months and periodically thereafter, as clinically appropriate, as well as over the first four to six weeks after any dose increase. In addition, monitoring of FPG 4 weeks and HbA_{1c} 3 months after the end of the treatment should be performed.

If hyperglycaemia develops in a patient being treated with Signifor LAR, the initiation or adjustment of antidiabetic treatment is recommended, following the established treatment guidelines for the management of hyperglycaemia. If uncontrolled hyperglycaemia persists despite appropriate medical management, the dose of Signifor LAR should be reduced or Signifor LAR treatment discontinued (see also section Interaction with other medicinal products and other forms of interaction).

There have been post-marketing cases of ketoacidosis with Signifor LAR in patients with and without a history of diabetes. Patients who present with signs and symptoms consistent with severe metabolic acidosis should be assessed for ketoacidosis regardless of diabetes history.

Patients with poor glycaemic control (as defined by HbA_{1c} values >8% while receiving anti-diabetic therapy), diabetes management and monitoring should be intensified prior to initiation and during pasireotide therapy.

Liver tests

Mild transient elevations in aminotransferases are commonly observed in patients treated with pasireotide. Rare cases of concurrent elevations in ALT (alanine aminotransferase) greater than 3 x ULN and bilirubin greater than 2 x ULN have also been observed (see section Undesirable effects). Monitoring of liver function is recommended prior to treatment with pasireotide intramuscular use and after the first two to three weeks, then monthly for three months on treatment. Thereafter liver function should be monitored as clinically indicated.

Patients who develop increased transaminase levels should be monitored frequently until values return to pre-treatment levels. Therapy with pasireotide should be discontinued if the patient develops jaundice or other signs suggestive of clinically significant liver dysfunction, in the event of a sustained increase in AST (aspartate aminotransferase) or ALT of 5 x ULN or greater, or if ALT or AST elevations greater than 3 x ULN occur concurrently with bilirubin elevations greater than 2 x ULN. Following discontinuation of treatment with pasireotide, patients should be monitored until resolution. Treatment should not be restarted if the liver function abnormalities are suspected to be related to pasireotide.

Cardiovascular related events

Bradycardia has been reported with the use of pasireotide (see section Undesirable effects). Careful monitoring is recommended in patients with cardiac disease and/or risk factors for bradycardia, such as history of clinically significant bradycardia or acute myocardial infarction, high-grade heart block, congestive heart failure (NYHA Class III or IV), unstable angina, sustained ventricular tachycardia, ventricular fibrillation. Dose adjustment of medicinal products such as beta blockers, calcium channel blockers, or medicinal products to control electrolyte balance, may be necessary (see also section Interaction with other medicinal products and other forms of interaction).

Pasireotide has been shown to prolong the QT interval on the ECG in two dedicated healthy volunteer studies performed with the subcutaneous formulation. The clinical significance of this prolongation is unknown. The phase III clinical studies in acromegaly patients did not identify any clinically meaningful differences in the QT prolongation events between pasireotide intramuscular use and the somatostatin analogues which were tested as active comparator. All QT-related events were transient and resolved without therapeutic intervention.

Episodes of torsade de pointes were not observed in any clinical study with pasireotide.

Pasireotide should be used with caution and the benefit risk carefully weighed in patients who are at significant risk of developing prolongation of QT, such as those:

- with congenital long QT syndrome.
- with uncontrolled or significant cardiac disease, including recent myocardial infarction, congestive heart failure, unstable angina or clinically significant bradycardia.
- taking antiarrhythmic medicinal products or other substances that are known to lead to QT prolongation (see section Interaction with other medicinal products and other forms of interaction).
- with hypokalaemia and/or hypomagnesaemia.

A baseline ECG is recommended prior to initiating therapy with Signifor LAR. Monitoring for an effect on the QTc interval is advisable 21 days after the beginning of the treatment and as clinically indicated thereafter. Hypokalaemia and/or hypomagnesaemia must be corrected prior to administration of Signifor LAR and should be monitored periodically during therapy.

Hypocortisolism

The suppression of ACTH (adrenocorticotrophic hormone) secretion can result in hypocortisolism in patients treated with Signifor LAR. It is therefore necessary to monitor and instruct patients on the signs and symptoms associated with hypocortisolism (e.g. weakness, fatigue, anorexia, nausea, vomiting, hypotension, hyperkalaemia, hyponatraemia, hypoglycaemia). In the event of documented hypocortisolism, temporary exogenous steroid (glucocorticoid) replacement therapy and/or dose reduction or interruption of Signifor LAR therapy may be necessary. Rapid decreases in cortisol levels may be associated with decreases in white blood cell count.

Gallbladder and related events

Cholelithiasis (gallstones) is a recognised adverse reaction associated with somatostatin analogues and has frequently been reported in clinical studies with pasireotide (see section Undesirable effects). There have been post-marketing cases of cholangitis in patients taking Signifor LAR, which in the majority of cases was reported as a complication of gallstones. Ultrasonic examination of the gallbladder before and at 6 to 12 month intervals during Signifor LAR therapy is therefore recommended. The presence of gallstones in Signifor LAR-treated patients is largely asymptomatic; symptomatic stones should be managed according to clinical practice.

Pituitary hormones

As the pharmacological activity pasireotide mimics that of somatostatin, inhibition of pituitary hormones other than GH and/or IGF-1 in patients with acromegaly and ACTH/cortisol in patients with disease cannot be ruled out. Monitoring of pituitary function (e.g. TSH/free T₄) before and periodically during Signifor LAR therapy should therefore be considered, as clinically appropriate.

Effect on female fertility

The therapeutic benefits of a reduction in growth hormone (GH) levels and normalisation of insulin like-growth factor 1 (IGF-1) concentration in female acromegalic patients could potentially restore fertility. Female patients of childbearing potential should be advised to use adequate contraception if necessary during treatment with Signifor LAR (see section Fertility, pregnancy and lactation).

Coagulation abnormalities

Patients with significantly increased prothrombin time (PT) and partial thromboplastin time (PTT) values or patients receiving coumarin-derivative or heparin-derivative anticoagulants were excluded from clinical studies with pasireotide as the safety of the combination with such anticoagulants has not been established. If concomitant use of coumarin-derivative or heparin-derivative anticoagulants with Signifor LAR intramuscular use cannot be avoided, patients should be monitored regularly for alterations in their coagulation parameters (PT and PTT) and the anticoagulant dose adjusted accordingly.

Renal impairment

Due to the increase in unbound drug exposure, Signifor LAR should be used with caution in patients with severe renal impairment or end stage renal disease (see section Pharmacokinetic properties).

Sodium content

This medicinal product contains less than 1 mmol (23 mg) sodium per dose i.e. it is essentially 'sodium-free'.

Interaction with other medicinal products and other forms of interaction

Anticipated pharmacokinetic interactions resulting in effects on pasireotide

The influence of the P-gp inhibitor verapamil on the pharmacokinetics of subcutaneous pasireotide was tested in a drug drug interaction study in healthy volunteers. No change in the pharmacokinetics (rate or extent of exposure) of pasireotide was observed.

Anticipated pharmacokinetic interactions resulting in effects on other medicinal products

Pasireotide may decrease the relative bioavailability of ciclosporin. Concomitant administration of pasireotide and ciclosporin may require adjustment of the ciclosporin dose to maintain therapeutic levels.

Anticipated pharmacodynamic interactions

Medicinal products that prolong the QT interval

Pasireotide should be used with caution in patients who are concomitantly receiving medicinal products that prolong the QT interval, such as class Ia antiarrhythmics (e.g. quinidine, procainamide, disopyramide), class III antiarrhythmics (e.g. amiodarone, dronedarone, sotalol, dofetilide, ibutilide), certain antibacterials (intravenous erythromycin, pentamidine injection, clarithromycin, moxifloxacin), certain antipsychotics (e.g. chlorpromazine, thioridazine, fluphenazine, pimozide, haloperidol, tiagride, amisulpride, sertindole, methadone), certain antihistamines (e.g. terfenadine, astemizole, mizolastine), antimalarials (e.g. chloroquine, halofantrine, lumefantrine), certain antifungals (ketoconazole, except in shampoo) (see also section Special warnings and precautions for use).

Bradycardic medicinal products

Clinical monitoring of heart rate, notably at the beginning of treatment, is recommended in patients receiving pasireotide concomitantly with bradycardic medicinal products, such as beta blockers (e.g. metoprolol, carteolol, propranolol, sotalol), acetylcholinesterase inhibitors (e.g. rivastigmine, physostigmine), certain calcium channel blockers (e.g. verapamil, diltiazem, bepridil), certain antiarrhythmics (see also section Special warnings and precautions for use).

Insulin and antidiabetic medicinal products

Dose adjustments (decrease or increase) of insulin and antidiabetic medicinal products (e.g. metformin, liraglutide, vildagliptin, nateglinide) may be required when administered concomitantly with pasireotide (see also section Special warnings and precautions for use).

Fertility, pregnancy and lactation

Pregnancy

There is a limited amount of data from the use of pasireotide in pregnant women. Studies in animals in which pasireotide was administered via the subcutaneous route have shown reproductive toxicity (see section Preclinical safety data). Pasireotide is not recommended for use during pregnancy and in women of childbearing potential who are not using contraception (see section Special warnings and precautions for use).

Breast-feeding

It is unknown whether pasireotide is excreted in human milk. Available data in rats in which pasireotide was administered via the subcutaneous route have shown excretion of pasireotide in milk (see section Preclinical safety data). Breast feeding should be discontinued during treatment with Signifor LAR.

Fertility

Studies in rats in which pasireotide was administered via the subcutaneous route have shown effects on female reproductive parameters (see section Preclinical safety data). The clinical relevance of these effects in humans is unknown.

Effects on ability to drive and use machines

Signifor LAR may have a minor influence on the ability to drive and use machines. Patients should be advised to be cautious when driving or using machines if they experience fatigue, dizziness or headache during treatment with Signifor LAR.

Undesirable effects

Summary of the safety profile

Safety assessment was made based on 491 acromegaly patients who received pasireotide (419 patients received pasireotide intramuscular use and 72 received pasireotide subcutaneous use) in phase I, II and III studies. The safety profile of pasireotide intramuscular use was consistent with the somatostatin analogue class, except for the higher degree and frequency of hyperglycaemia seen with pasireotide intramuscular use.

The most common adverse reactions (incidence ≥1/10) from the pooled safety data from the phase III studies C2305 and C2402 were (in decreasing order): diarrhoea (most common in study C2305), cholelithiasis, hyperglycaemia (most common in study C2402) and diabetes mellitus. Common Toxicity Criteria (CTC) Grade 3 and 4 adverse reactions were mostly related to hyperglycaemia.

Tabulated list of adverse reactions

The adverse reactions in Table 1 include events reported in the pivotal studies with the intramuscular formulation in patients with acromegaly. Adverse reactions are listed according to MedDRA primary system organ class. Within each system organ class, adverse reactions are ranked by frequency. Within each frequency grouping, adverse reactions are presented in the order of decreasing seriousness. Frequencies were defined as follows: Very common (≥1/10); common (≥1/100 to <1/10); uncommon (≥1/1,000 to <1/100); not known (cannot be estimated from the available data)

Table 1 Adverse reactions by preferred term for pasireotide intramuscular use

System Organ Class	Very common (≥1/10)	Common (≥1/100 to <1/10)	Uncommon (≥1/1,000 to <1/100)	Not known
Blood and lymphatic system disorders		Anaemia		
Endocrine disorders		Adrenal insufficiency*		
Metabolism and nutrition disorders	Hyperglycaemia, diabetes mellitus	Type 2 diabetes mellitus, glucose tolerance impaired, decreased appetite		Diabetic ketoacidosis
Nervous system disorders		Headache, dizziness		
Cardiac disorders		Sinus bradycardia**, QT prolongation		
Gastrointestinal disorders	Diarrhoea, nausea, abdominal pain*	Abdominal distension, vomiting		Steatorrhea Faeces discoloured
Hepatobiliary disorders	Cholelithiasis	Cholecystitis*, cholelasis		
Skin and subcutaneous tissue disorders		Alopecia, pruritus		
General disorders and administration site conditions	Fatigue*	Injection site reaction***		
Investigations		Glycosylated haemoglobin increased, alanine aminotransferase increased, aspartate aminotransferase increased, gamma-glutamyltransferase increased, blood glucose increased, blood creatinine phosphokinase increased, lipase increased	Amylase increased, prothrombin time prolonged	

* Grouped terms: Adrenal insufficiency includes adrenal insufficiency and blood cortisol decreased.

** Sinus bradycardia includes bradycardia and sinus bradycardia. Abdominal pain includes abdominal pain and abdominal pain upper.

*** Injection site reaction includes injection site pain, injection site nodule, injection site discomfort, injection site bruising, injection site pruritus, injection site reaction, injection site hypersensitivity and injection site swelling. Cholecystitis includes cholecystitis acute and cholecystitis chronic. Fatigue includes fatigue and asthenia.

Description of selected adverse reactions

Glucose metabolism disorders

Elevated fasting glucose level was the most frequently reported grade 3/4 laboratory abnormality in the two phase III studies. In study C2305, grade 3 elevated fasting glucose levels were reported in 9.7% and 0.6% and grade 4 in 0.6% and 0% of acromegaly patients treated with pasireotide intramuscular use and octreotide intramuscular use, respectively. In study C2402, grade 3 elevated fasting glucose levels were reported in 14.3% and 17.7% of acromegaly patients treated with pasireotide intramuscular use 40 mg and 60 mg respectively, and in no patients in the active control group. Two cases of hyperglycaemia related emergencies (diabetic ketoacidosis and diabetic hyperglycaemic coma) were reported following a dose increase of pasireotide to 60 mg in medical treatment naive patients; one in a patient with untreated hyperglycaemia and HbA_{1c} >8% prior to initiation of pasireotide and the other in a patient with untreated hyperglycaemia and a fasting plasma glucose of 339 mg/dl, respectively. In both studies, mean FPG and HbA_{1c} levels peaked within the first three months of treatment with pasireotide intramuscular use. In medically naive patients (study C2305), the mean absolute increase in FPG and HbA_{1c} was similar at most of the time points for all patients treated with pasireotide intramuscular use irrespective of baseline values.

The degree and frequency of hyperglycaemia observed in the two pivotal studies in acromegaly patients were higher with Signifor LAR intramuscular use than with active control (octreotide intramuscular use or lanreotide deep subcutaneous injection). In a pooled analysis of the two pivotal studies, the overall incidence of hyperglycaemia-related adverse reactions was 58.6% (all grades) and 9.9% (CTC Grade 3 and 4) for Signifor LAR intramuscular use versus 18.0% (all grades) and 1.1% (CTC Grade 3 and 4) for the active control. In the pivotal study with patients inadequately controlled on another somatostatin analogue, the proportion of patients not previously treated with anti-diabetic agents who required commencement of anti-diabetic therapy during the study was 17.5% and 16.1% in the Signifor LAR 40 mg and 60 mg arms compared to 1.5% in the active control arm. In the pivotal study with patients who did not receive prior medical treatment, the proportion of patients who required commencement of anti-diabetic therapy during the study was 36% in the Signifor LAR arm compared to 4.4% in the active control arm.

The elevations of fasting plasma glucose and HbA_{1c} observed with pasireotide intramuscular use treatment are reversible after discontinuation.

Monitoring of blood glucose levels in patients treated with Signifor LAR is recommended (see section Special warnings and precautions for use).

Gastrointestinal disorders

Gastrointestinal disorders were frequently reported with Signifor LAR. These reactions were usually of low grade, required no intervention and improved with continued treatment. Gastrointestinal disorders were less frequent in inadequately controlled patients compared to medically naive patients.

Injection site reactions

In the phase III studies, injection site related reactions (e.g. injection site pain, injection site discomfort) were mostly grade 1 or 2 in severity. The incidence of such events was highest in the first 3 months of treatment. In the acromegaly studies, the events were comparable between pasireotide intramuscular use and octreotide intramuscular use treated patients, and were less frequent in inadequately controlled patients compared to medically naive patients.

QT prolongation

In study C2305 the proportion of patients with newly occurring notable QT/QTc intervals was comparable between pasireotide intramuscular use and octreotide intramuscular use groups up to crossover, with few notable outlying values. QTcF >480 ms was reported for 3 versus 2 patients in the pasireotide intramuscular use and octreotide intramuscular use groups, respectively, and QTcF >60 ms prolonged from baseline was reported for 2 versus 1 patients in the respective groups. In study C2402, the only notable outlier was a QTcF value >480 ms in 1 patient in the pasireotide intramuscular use 40 mg group. No QTcF values >500 ms were observed in any of the pivotal studies.

Liver enzymes

Transient elevations in liver enzymes have been reported with the use of somatostatin analogues and were also observed in healthy subjects and patients receiving pasireotide in clinical studies. The elevations were mostly asymptomatic, of low grade and reversible with continued treatment. A few cases of concurrent elevations in ALT greater than 3 x ULN and bilirubin greater than 2 x ULN have been observed with the subcutaneous formulation, however not in patients treated with pasireotide intramuscular use. All observed cases of concurrent elevations were identified within ten days of initiation of treatment. The patients recovered without clinical sequelae and liver function test results returned to baseline values after discontinuation of treatment.

Monitoring of liver enzymes is recommended before and during treatment with Signifor LAR (see section Special warnings and precautions for use), as clinically appropriate.

Pancreatic enzymes

Asymptomatic elevations in lipase and amylase were observed in patients receiving pasireotide in clinical studies. The elevations were mostly low grade and reversible while continuing treatment. Pancreatitis is a potential adverse reaction associated with the use of somatostatin analogues due to the association between cholelithiasis and acute pancreatitis.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

Overdose

In the event of overdose, it is recommended that appropriate supportive treatment be initiated, as dictated by the patient's clinical status, until resolution of the symptoms.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Pituitary and hypothalamic hormones and analogues, somatostatin and analogues, ATC code: H01CB05

Mechanism of action

Pasireotide is a cyclohexapeptide, injectable somatostatin analogue. Like the natural peptide hormones somatostatin-14 and somatostatin-28 (also known as somatotropin release inhibiting factor [SRIF]) and other somatostatin analogues, pasireotide exerts its pharmacological activity via binding to somatostatin receptors. Five human somatostatin receptor subtypes are known: hsst1, 2, 3, 4, and 5. These receptor subtypes are expressed in different tissues under normal physiological conditions. Somatostatin analogues bind to hsst receptors with different potencies (see Table 2). Pasireotide binds with high affinity to four of the five hssTs.

Table 2 Binding affinities of somatostatin (SRIF-14), pasireotide, octreotide and lanreotide to the five human somatostatin receptor subtypes (hsst1-5)

Compound	hsst1	hsst2	hsst3	hsst4	hsst5
Somatostatin (SRIF-14)	0.93±0.12	0.15±0.02	0.56±0.17	1.5±0.4	0.29±0.04
Pasireotide	9.3±0.1	1.0±0.1	1.5±0.3	>1,000	0.16±0.01
Octreotide	280±80	0.38±0.08	7.1±1.4	>1,000	6.3±1.0
Lanreotide	180±20	0.54±0.08	14±9	230±40	17±5

Results are the mean±SEM of IC₅₀ values expressed as nmol/l.

Pharmacodynamic effects

Somatostatin receptors are expressed in many tissues, especially in neuroendocrine tumours in which hormones are excessively secreted, including GH in acromegaly.

Due to its broad binding profile to somatostatin receptors, pasireotide has the potential to stimulate both hsst2 and hsst5 subtype receptors relevant for inhibition of GH and IGF-1 secretion and therefore to be effective for the treatment of acromegaly.

Glucose metabolism

In a randomised double-blinded mechanism study conducted in healthy volunteers, the development of hyperglycaemia with pasireotide administered as pasireotide subcutaneous use at doses of 0.6 and 0.9 mg twice a day was related to significant decreases in insulin secretion as well as incretin hormones (i.e. glucagon-like peptide-1 [GLP-1] and glucose-dependent insulinotropic polypeptide [GIP]). Pasireotide did not affect insulin sensitivity.

Clinical efficacy and safety

The efficacy of pasireotide intramuscular use has been demonstrated in two phase III, multicentre studies.

Study C2402, inadequately controlled patients

Study C2402 was a phase III, multicentre, randomised, parallel group, three arm study of double- blind pasireotide intramuscular use 40 mg and 60 mg versus open-label octreotide intramuscular use 30 mg or lanreotide deep subcutaneous injection 120 mg in patients with inadequately controlled acromegaly. A total of 198 patients were randomised to receive pasireotide intramuscular use 40 mg (n=65), pasireotide intramuscular use 60 mg (n=65) or active control (n=68). 192 patients were treated. A total of 181 patients completed the core phase (24 weeks) of the study.

Inadequately controlled patients in study C2402 are defined as patients with a mean GH concentration of a 5-point profile over a 2-hour period >2.5 µg/l and sex- and age-adjusted IGF-1 >1.3 x ULN. Patients had to be treated with maximum indicated doses of octreotide intramuscular use (30 mg) or lanreotide deep subcutaneous injection (120 mg) for at least 6 months prior to randomisation. Three-quarters of patients had previously been treated with octreotide intramuscular use and a quarter with lanreotide deep subcutaneous injection. Nearly half of the patients had additional prior medical treatment for acromegaly other than somatostatin analogues. Two-thirds of all patients had undergone prior surgery. Baseline mean GH was 17.6 µg/l, 12.1 µg/l and 9.5 µg/l in the 40 mg, 60 mg and active control groups, respectively IGF-1 mean values at baseline were 2.6, 2.8 and 2.9 x ULN, respectively.

The primary efficacy endpoint was to compare the proportion of patients achieving biochemical control (defined as mean GH levels <2.5 µg/l and normalisation of sex- and age-adjusted IGF-1 at week 24 with pasireotide intramuscular use 40 mg or 60 mg versus continued treatment with active control (octreotide intramuscular use 30 mg or lanreotide deep subcutaneous injection 120 mg), separately. The study met its primary efficacy endpoint for both pasireotide intramuscular use doses. The proportion of patients achieving biochemical control was 15.4% (p-value = 0.0006) and 20.0% (p-value <0.0001) for pasireotide intramuscular use 40 mg and 60 mg, respectively at 24 weeks compared with zero in the active control arm (Table 3).

Table 3 Key results at week 24 (Study C2402)

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Based on *in vitro* data pasireotide appears to be a substrate of efflux transporter P-gp (P-glycoprotein). Based on *in vitro* data pasireotide is not a substrate of the efflux transporter BCRP (breast cancer resistance protein) nor of the influx transporters OCT1 (organic cation transporter 1), OATP (organic anion-transporting polypeptide) 1B1, 1B3 or 2B1. At therapeutic dose levels pasireotide is also not an inhibitor of UGT1A1, OATP1B1 or 1B3, OAT1 or OAT3, OCT1 or OCT2, P-gp, BCRP, MRP2 and BSEP.

Biotransformation

Pasireotide is metabolically highly stable and *in vitro* data show that pasireotide is not a substrate, inhibitor or inducer of CYP450. In healthy volunteers, pasireotide is predominantly found in unchanged form in plasma, urine and faeces.

Elimination

Pasireotide is eliminated mainly via hepatic clearance (biliary excretion), with a small contribution of the renal route. In a human ADME study 55.9±6.63% of the radioactive pasireotide subcutaneous dose was recovered over the first 10 days after administration, including 48.3±8.16% of the radioactivity in faeces and 7.63±2.03% in urine.

The apparent clearance (CL/F) of pasireotide intramuscular use in healthy volunteers is on average 4.5-8.5 litres/h. Based on population pharmacokinetic (PK) analyses, the estimated CL/F was approximately 5.6 to 8.2 litres/h for typical acromegaly patients.

Linearity and time dependency

Pharmacokinetic steady state for pasireotide intramuscular use is achieved after three months. Following multiple monthly doses, pasireotide intramuscular use demonstrates approximately dose-proportional pharmacokinetic exposures in the dose range of 10 mg to 60 mg every 4 weeks.

Special populations

Paediatric population

No studies have been performed in paediatric patients.

Patients with renal impairment

Renal clearance has a minor contribution to the elimination of pasireotide in humans. In a clinical study with single subcutaneous dose administration of 900 µg pasireotide in subjects with impaired renal function, renal impairment of mild, moderate or severe degree, or end stage renal disease (ESRD) did not have a significant impact on total pasireotide plasma exposure. The unbound plasma pasireotide exposure (AUC₀₋₂₄) was increased in subjects with renal impairment (mild: 33%; moderate: 25%; severe: 99%; ESRD: 143%) compared to control subjects.

Patients with hepatic impairment

No clinical studies in subjects with liver impairment have been performed with pasireotide intramuscular use. In a clinical study of a single subcutaneous dose of pasireotide in subjects with impaired hepatic function, statistically significant differences were found in subjects with moderate and severe hepatic impairment (Child-Pugh B and C). In subjects with moderate and severe hepatic impairment, AUC₀₋₂₄ was increased 60% and 79%, C_{max} was increased 67% and 69%, and CL/F was decreased 37% and 44%, respectively.

Elderly patients (≥65 years)

Age is not a significant covariate in the population pharmacokinetic analysis of patients with acromegaly.

Demographics

Population PK analyses of pasireotide intramuscular use suggest that race does not influence PK parameters. PK exposures had a slight correlation with body weight in the study with medical treatment naive patients, but not in the study with inadequately controlled patients. Female acromegaly patients had a higher exposure of 32% and 51% compared to male patients in studies with medical treatment naive patients and inadequately controlled patients, respectively; these differences in exposure were not clinically relevant based on efficacy and safety data.

Preclinical safety data

Non-clinical safety data from studies performed with pasireotide administered via the subcutaneous route reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity and carcinogenic potential. Additionally, tolerability and repeated dose toxicity studies were conducted with pasireotide via the intramuscular route. Most findings seen in repeated toxicity studies were reversible and attributed to the pharmacology of pasireotide. Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use.

Pasireotide administered via the subcutaneous route did not affect fertility in male rats but, as expected from the pharmacology of pasireotide, females presented abnormal cycles or acyclicity, and decreased numbers of corpora lutea and implantation sites. Embryo toxicity was seen in rats and rabbits at doses that caused maternal toxicity but no teratogenic potential was detected. In the pre- and postnatal study in rats, pasireotide had no effects on labour and delivery, but caused slight retardation in the development of pinna detachment and reduced body weight of the offspring.

Available toxicological data in animals have shown excretion of pasireotide in milk.

PHARMACEUTICAL PARTICULARS

List of excipients

Powder

Poly(D,L-lactide-co-glycolide) (50 60:40 50)

Poly(D,L-lactide-co-glycolide) (50:50)

Solvent

Carmellose sodium

Mannitol

Poloxamer 188

Water for injections

Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products

STORAGE

See folding box.

Store in a refrigerator (2 to 8°C)

Do not freeze.

PACK SIZE

A box of 1 vial of powder of suspension for injection and 1 pre-filled syringe with solvent, co-packaged in a sealed blister tray with one vial adapter and one safety-engineered needle for injection.

INSTRUCTIONS FOR USE

Instructions for preparation and intramuscular injection of Signifor LAR

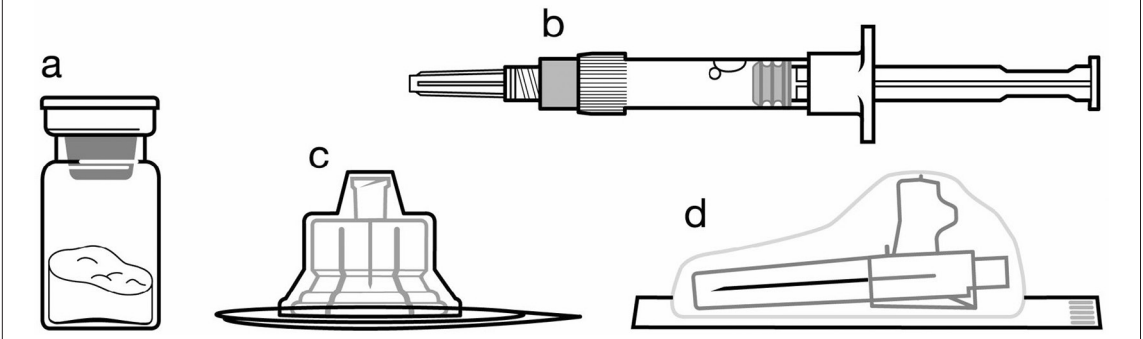
FOR DEEP INTRAMUSCULAR INJECTION ONLY

ATTENTION:

There are 2 critical steps in the reconstitution of Signifor LAR. **Not Following them could result in failure to deliver the drug appropriately.**

- The injection kit must reach room temperature. Remove the injection kit from the fridge and let the kit stand at room temperature for a minimum of 30 minutes before reconstitution, but not exceed 24 hours.
- After adding the diluent solution, **shake the vial moderately** in a horizontal direction for a minimum of 30 seconds **until uniform suspension is formed.**

Included in the injection kit:



- a One vial containing Signifor LAR powder
- b One prefilled syringe containing the diluent solution for reconstitution
- c One vial adapter for drug product reconstitution
- d One safety injection needle (20G x 1.5")

Follow the instructions below carefully to ensure proper reconstitution of Signifor LAR before deep intramuscular injection.

Signifor LAR suspension must only be prepared **immediately** before administration.

Signifor LAR should only be administered by a trained health professional.

Step 1

Remove the Signifor LAR injection kit from refrigerated storage.

ATTENTION: It is essential to start the reconstitution process only after the injection kit reaches room temperature. Let the kit stand at room temperature for a minimum of 30 minutes before reconstitution, but do not exceed 24 hours.

Note: The injection kit can be re-refrigerated if needed.



Step 2

Remove the plastic cap from the vial and clean the rubber stopper of the vial with an alcohol wipe.



Remove the lid film of the vial adapter packaging, but do NOT remove the vial adapter from its packaging.

Holding the vial adapter packaging, position the vial adapter on top of the vial and push it fully down so that it snaps in place, confirmed by an audible “click”.

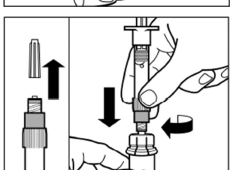


Lift the packaging off the vial adapter with a vertical movement.

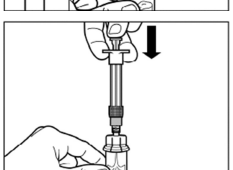


Step 3

Remove the cap from the syringe prefilled with diluent solution and **screw** the syringe onto the vial adapter.

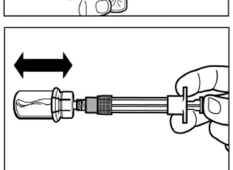


Slowly push the plunger all the way down to transfer all the diluent solution in the vial.



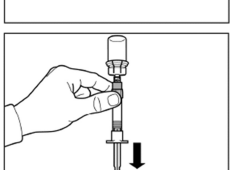
Step 4

ATTENTION: Keep the plunger pressed and shake the vial **moderately** in a horizontal direction for a minimum of 30 seconds so that the powder is completely suspended. **Repeat moderate shaking for another 30 seconds if the powder is not completely suspended.**

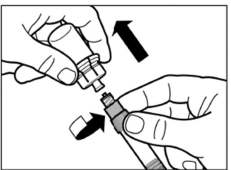


Step 5

Turn syringe and vial upside down, **slowly** pull the plunger back and draw the entire content from the vial into the syringe.

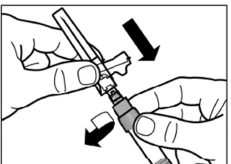


Unscrew the syringe from the vial adapter.



Step 6

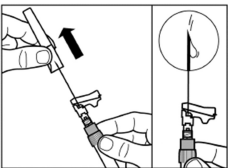
Screw the safety injection needle onto the syringe.



Pull the protective cover straight off the needle.

To avoid sedimentation, you may gently shake the syringe to maintain a uniform suspension.

Gently tap the syringe to remove any visible bubbles and expel them from the syringe.



The reconstituted Signifor LAR is now ready for **immediate** administration.

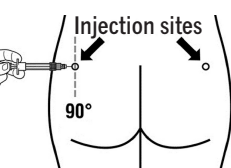
Step 7

Signifor LAR must be given only by deep intramuscular injection; **NEVER** intravenously.

Prepare the injection site with an alcohol wipe.

Insert the needle fully into the left or right gluteus at a 90° angle to the skin.

Slowly pull back the plunger to check that no blood vessel has been penetrated (reposition if a blood vessel has been penetrated).

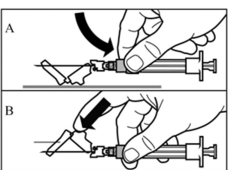


Slowly depress the plunger until the syringe is empty. Withdraw the needle from the injection site and activate the safety guard (as shown in Step 8).

Step 8

Activate the safety guard over the needle, in one of the 2 methods shown:

- either press the hinged section of the safety guard down onto a hard surface (figure A).
- or push the hinge forward with your finger (figure B).



An audible “click” confirms proper activation.

Dispose of syringe immediately in a sharps container.



Product Registration Holder:

FTA Pharma Sdn. Bhd.
C-9-13A, C-9-16 & C-9-35, Centum@Oasis Corporate Park
Jalan PJU1A/2 Ara Damansara
47301 Petaling Jaya,
Selangor, Malaysia.

Malaysia Package Insert

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®=registered trademark

Manufacturer:

Recordati AG Rare Diseases Branch
Lichstrasse 35
4056 Basel
Switzerland